Toroid Models for Quantum-Gravitational Resonance via de Broglie–Schwarzschild Symmetry

Version 4 (corrected)

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July 23, 2025 (revised June 2026; corrected October 2026)

Version 4 (corrected, October 2026)
Zenodo DOI: 10.5281/zenodo.23811063

Abstract

We propose a new resonance condition in which the de Broglie wavelength of a quantum system matches the Schwarzschild radius of its gravitational mass:

$\lambda = R_s$.

In a single body this relationship arises at the Planck scale under relativistic velocities, and, as later shown, also for larger masses at very low speeds. Here we suggest a two-subsystem design that may bring it within experimental reach.

In this framework, gravitational mass and quantum wave behavior are separated into two interacting subsystems composed of known materials: tungsten and graphene. Tungsten provides the dominant rest mass (with $R_s = \frac{2GM}{c^2}$), while conduction electrons in the graphene supply tunable quantum waves (with $\lambda = \frac{h}{mv}$).

We examine as the primary configuration:

1. A graphene-coated tungsten toroid.

As further avenues for research we propose:

2. Graphene nanoparticle clouds around tungsten core toroid. 3. Nanoparticle ensembles of pure tungsten and graphene-coated tungsten nanoparticles.

These configurations are explored as a route toward the condition $\lambda = R_s \approx 10^{-26}\,\mathrm{m}$, requiring no exotic particles or modifications to known physics. Version 4 states the main practical limitation of the graphene design (Section 8.6). This framework thus suggests a new route for exploring quantum–gravitational interaction in experimentally accessible regimes.

The author uses the pseudonym Gravity Resonance to preserve anonymity and avoid personal indexing, while welcoming collaboration via Zenodo private messages.

Corrections in Version 4 (October 2026). (1) References added to Carr's earlier work. (2) A single body also satisfies $\lambda = R_s$: at the Planck mass at $v \approx 0.95\,c$, and heavier bodies at lower speeds (later shown in A5, to be developed further). (3) Tables, figures and several numbers corrected; the main limitation of the graphene design is stated. (4) New Section 11: an update with matching speeds for any mass. The idea, structure and style of the paper are otherwise unchanged.

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1 Introduction

The gulf between quantum mechanics and general relativity has long appeared insurmountable, separated by extreme energy scales and conceptual divides. Yet the fundamental expressions for a quantum wavelength and a gravitational radius suggest a surprising symmetry:

$$\lambda = \frac{h}{mv} \quad \text{(de Broglie wavelength)} \tag{1}$$

$$R_s = \frac{2GM}{c^2} \quad \text{(Schwarzschild radius)} \tag{2}$$

These expressions depend only on mass and velocity—quantities accessible in a laboratory. What if we could design a system in which these two characteristic lengths become equal: $\lambda = R_s$? Could such a configuration expose an underlying resonance between quantum and gravitational domains?

In this paper, we investigate the conditions under which the de Broglie wavelength of a quantum system can match the Schwarzschild radius of its gravitational mass.

We focus on a primary realizable configuration: a graphene-coated tungsten toroid, in which conduction electrons in the graphene layer form a tunable wave-like subsystem while the tungsten core provides the dominant mass.

As a conceptual extension, Appendix B discusses alternative pathways involving nanoparticles—such as clouds of tungsten nanoparticles coated with graphene, or mixed arrangements of pure tungsten and graphene-coated particles. Future avenues may also include configurations with a massive tungsten core toroid surrounded by graphene nanoparticles, or other combinations in which the total number of conduction electrons exceeds that in the original graphene sheet. Such setups could, in principle, allow the de Broglie wavelength λ to match even smaller Schwarzschild radii R_s more easily, while remaining a speculative research direction.

In the toroid configuration, the electron velocity v can be tuned via applied voltages or currents, allowing the de Broglie wavelength to approach the Schwarzschild radius. The resonance is geometric and based on the material properties and electron motion, without invoking exotic fields, extra dimensions, or a full quantum gravity theory.

The resonance condition $\lambda = R_s$ is motivated not only by symmetry but also by historical and theoretical hints—ranging from de Broglie's matter-wave hypothesis to recent studies of the Compton–Schwarzschild correspondence [3]. It may be that the gravitational radius of a system and the wavelength of its internal quanta are not merely coincidental features—but co-defining properties of a deeper unification.

This initial article concentrates on the basic principle of the sought equivalence between the de Broglie wavelength (λ) and the Schwarzschild radius (R_s), establishing a theoretical foundation for their potential resonance. The details of the concrete model—a graphene-coated tungsten toroid —is presented not due to proficiency in technical intricacies, but to illustrate how such a symmetry might be practically explored. I invite collaborators to refine these models and propose novel configurations, fostering a collective effort to advance this quantum-gravitational inquiry.

I publish this work under the pseudonym **Gravity Resonance** to protect personal anonymity while inviting open and serious collaboration. This idea has been independently developed over decades, predating AI tools, and is now offered to the scientific community for exploration, refinement, and—most importantly—experimental testing.

2 Conceptual Background: The Planck Threshold

The Schwarzschild radius represents the gravitational curvature exerted by any mass in spacetime, independent of black hole collapse. We explore whether this subtle curvature may interact with the de Broglie wavelength of a quantum subsystem – the only known wavelength scale that approaches 10^{-26} to 10^{-35} meters.

The central idea:

- The toroid mass contributes R_s.
- A flowing or oscillating carrier system contributes λ.

These may be brought into resonance to investigate spacetime topology at mesoscopic scales. Using the relations from Equations 1 and 2, we note that at the Planck mass $M_P \approx 2.18 \times 10^{-8}\,\mathrm{kg}$ they align near the speed of light (exactly at $\gamma v = \pi c$, $v \approx 0.953\,c$). Heavier bodies align at lower speeds, as later shown in A5 [9] (see Section 11). The proposed workaround is to engineer near-resonance by decoupling mass and velocity.

3 Historical Notes and Scientific Context

This inquiry began not from formal academic research, but from a long-standing personal immersion in astronomy and theoretical physics. Though my graduate studies were in geography, I had already been deeply engaged with astrophysics, quantum mechanics, and cosmology – both through independent study and public lectures I gave to students. I had considered pursuing a physics major and had been active as an amateur astronomer for years.

During summer astronomy camps, we would estimate the mass of the universe just for fun – using its radius and the Schwarzschild formula. To our surprise, the result came remarkably close to accepted values. That early exercise left a lasting impression on me: that deep physical truths can be found through simple equations, if one asks the right questions.

Between 1997 and 2002, I began comparing the Schwarzschild radius with various quantum scales. Initially, I focused on the Compton wavelength. Later, the de Broglie wavelength proved more suitable, since it depends on velocity and can be adjusted experimentally. I observed a striking result: for an object with Planck mass moving near the speed of light (at about $0.95\,c$), its Schwarzschild radius equals its de Broglie wavelength – a unique symmetry where quantum mechanics and general relativity intersect.

This resonance seemed so elegant and natural that I expected it to be widely known. But I found few, if any, references to it in the literature. I tried to raise the idea through scientific forums, letters to institutions such as Brookhaven and Fermilab, and personal outreach – but the concept often went unnoticed. Perhaps it was unfamiliar, or simply outside the usual research priorities.

Years later I learned how much earlier work existed on this link. In 1979 Carr and Rees placed all physical objects on one mass–size chart [4], where the quantum and

gravitational lines cross at the Planck scale. Carr later developed this into the Black Hole Uncertainty Principle correspondence [7, 8], and Lake and Carr proposed a Compton–Schwarzschild correspondence [3]. Gantert discussed a related Schwarzschild–de Broglie relation in a 2014 book [13]. My own comparison of the smallest black hole with the de Broglie wavelength was developed independently. Together these works show that a direct link between quantum wavelength and gravitational radius is a serious subject of research.

At the time I developed this idea, I had also reached out to institutions like Fermilab and Brookhaven with thoughts on this potential resonance – but received no response. Whether the lack of engagement was due to the unconventional framing, informal channel of communication, or institutional filtering is hard to determine.

Still, such episodes highlight a deeper issue in the scientific process. When promising ideas are left unexplored – not for lack of merit, but due to perceived threat, unfamiliarity, or hierarchy – science risks delaying discoveries that might serve humanity. History shows that openness can be decisive: Einstein's mass–energy relation, once accepted, transformed not only theoretical physics but modern technology – from nuclear power to GPS. Had that insight been silenced or lost, the trajectory of global progress could have been very different.

We cannot rewrite the past. But we shape the future by how we ask, explore, and share today. It is not only historians who make history. Scientists do too – every time they choose to look openly at the equations in front of them.

4 Rotation and Charge: Exploring Additional Coupling Effects

In addition to its mass, the toroidal system can be designed to rotate and carry electric charge. These features allow for the exploration of how angular momentum and electromagnetic interactions may couple with spacetime curvature in a controlled, mesoscopic environment.

Rotation introduces dynamic curvature effects, potentially simulating frame-dependent interactions or modifying the local geometry experienced by the quantum subsystem. Charged systems may also interact with vacuum fluctuations and test configurations involving electromagnetic-gravitational interplay.

While this model does not draw on specific relativistic metrics, such as those used in black hole solutions, it does propose a practical platform for probing whether rotational and charge-related effects influence quantum-gravitational resonance.

The conceptual foundation aligns with broader investigations into how rotation and charge affect wave amplification and coherence in gravitational fields. Notably, studies such as the recent black hole bomb experiments using electromagnetic systems [1, 2] provide relevant theoretical background, even if realized under different regimes. Furthermore, lattice models have explored thermalization at synthetic horizons [14].

This section emphasizes the flexibility of the toroidal setup: by tuning rotation rate, charge distribution, and current direction, one may create conditions that test not only $\lambda = R_s$ resonance, but also the role of symmetry, dynamics, and fields in gravitational coupling at mesoscopic scales.

5 Feasibility of Laboratory Resonance Between λ and R_s

5.1 Toroid Geometry with Graphene-Coated Tungsten

The original model consists of a compact toroidal tungsten core surrounded by a conductive shell of graphene. The heavy tungsten core contributes most of the system's gravitational mass, defining R_s, while the electrons in the graphene layer form the wave-like subsystem, whose de Broglie wavelength λ is tunable via applied voltage or current.

We explore conditions under which $\lambda = R_s$ can be achieved for electrons traveling at drift or relativistic velocities.

We explore here two resonance targets:

- Planck length: $\lambda = 1.616 \times 10^{-35}$ m
- Mesoscopic scale: $\lambda = 1 \times 10^{-26}$ m

Using the relation $\lambda = \frac{h}{mv}$, we estimate the mass–velocity combinations required for different materials. This allows evaluation of whether the target wavelength can be achieved in practice.

5.2 Summary Table: Theoretical Combinations

Medium	v **(m/s)**	**M** ($\lambda \approx 10^{-26}$ **m**)	R_s **(m)**	**M** ($\lambda \approx 10^{-35}$ **m**)	R_s **(m)**
Supercurrent	1	6.63×10^{-8}	9.83×10^{-35}	41.0	6.08×10^{-26}
Crystal	10^3	6.63×10^{-11}	9.83×10^{-38}	0.041	6.08×10^{-29}
Graphene	10^6	6.63×10^{-14}	9.83×10^{-41}	4.1×10^{-5}	6.08×10^{-32}
Argon Plasma	10^2	6.63×10^{-10}	9.83×10^{-37}	0.41	6.08×10^{-28}

Table 1: Mass (M, kg) and velocity (v, m/s) combinations for reaching target de Broglie wavelengths and Schwarzschild radii. R_s calculated via $R_s = \frac{2GM}{c^2}$ from the M in the same row.

5.3 Material Notes

- **Supercurrent**: Excellent coherence; however, very low velocities demand large mass ($\sim$ 41 kg) to approach Planck resonance.
- **Crystal**: Macroscopically stable; phonon coherence possible, but actual mobile mass is small.
- **Graphene**: High mobility (Fermi velocity $\sim 10^6$ m/s); offers compact, scalable systems. Promising for miniaturized toroids.
- **Argon Plasma**: Velocity tunable (10^2–10^3 m/s); challenging to stabilize.

This table serves as a first estimate. The total system feasibility depends not only on mass and speed but also coherence, confinement, and environmental stability. Some of these configurations could be optimized in high-vacuum, cryogenic, or layered lattice environments.

6 Relativistic Scenarios and Planck-Scale Conditions

While the previous table focused on mesoscopic, non-relativistic conditions for approximating $\lambda \approx R_s$, there also exists a unique relativistic boundary where quantum wavelength and gravitational curvature align precisely: the Planck scale.

We now examine relativistic λ and corresponding R_s for select cases at high velocities. The relativistic de Broglie wavelength is calculated via:

$$\lambda = \frac{h}{\gamma m v}, \quad \text{with} \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{3}$$

6.1 Relativistic Estimates for Matching $\lambda = R_s$

Material/Case	**Speed v (m/s)**	**Mass m (kg)**	$\lambda = \frac{h}{\gamma m v}$ (m)	$R_s = \frac{2Gm}{c^2}$ (m)
Planck Particle (Ideal)	2.86×10^8	2.176×10^{-8}	3.23×10^{-35}	3.23×10^{-35}
Relativistic Graphene†	1×10^6	4.1×10^{-2}	1.62×10^{-38}	6.08×10^{-29}
Relativistic Argon Plasma†	3×10^7	4.1×10^{-2}	5.36×10^{-40}	6.08×10^{-29}
Supercurrent (High mass)	1×10^3	67.3	9.84×10^{-39}	9.99×10^{-26}

Table 2: Relativistic de Broglie wavelengths and Schwarzschild radii for several hypothetical single-object configurations. The Planck-particle row is set to the exact match, $\gamma v = \pi c$ (corrected in Version 4). The other rows, with masses not tuned to the match, illustrate the scale separation that arises when wave behavior and curvature originate from the same body. A single body can also be tuned to $\lambda = R_s$ at $v = \pi M_P^2 c/m^2$ (Section 2); this paper instead separates mass and carriers into two subsystems (tungsten core + graphene carriers).

† *These entries assume unified systems where wave behavior and gravitational curvature originate from the same body. The actual experimental model separates mass and carrier subsystems (e.g., tungsten + graphene).*

Interpretation and Limitations

The relativistic estimates presented in Table 2 are provided to illustrate conditions under which a single object may simultaneously satisfy both $\lambda = R_s$ and relativistic de Broglie wavelength scaling. Each row assumes a unified mass system—that is, one body whose gravitational field and wave behavior arise from the same rest mass.

This contrasts with the primary experimental model discussed in this paper, where mass (M) and quantum carriers (m) are separated across two subsystems (e.g., a tungsten toroid and a graphene coating). Therefore, the values shown here are not meant to represent feasible lab configurations but rather theoretical symmetry points.

Note: The "Relativistic Graphene" row assumes a macroscopic graphene body of 0.041 kg acting as both the gravitational and quantum subsystem. This is purely hypothetical and does not reflect the actual graphene coating used in the toroid resonance model.

7 Graphical Analysis of Wavelength Matching

The following plot visualizes the de Broglie wavelength $\lambda = \frac{h}{mv}$ against velocity v for an idealized Planck-mass particle. Overlaid are two target wavelengths: the mesoscopic resonance scale ($\lambda = 10^{-26}$ m) and the Planck length ($\lambda = 1.616 \times 10^{-35}$ m). The curve uses the relativistic wavelength, Eq. (3), and stops below c. It meets the Planck-mass Schwarzschild radius, $R_s = 3.23 \times 10^{-35}$ m, at $v \approx 2.86 \times 10^8$ m/s (corrected in Version 4).

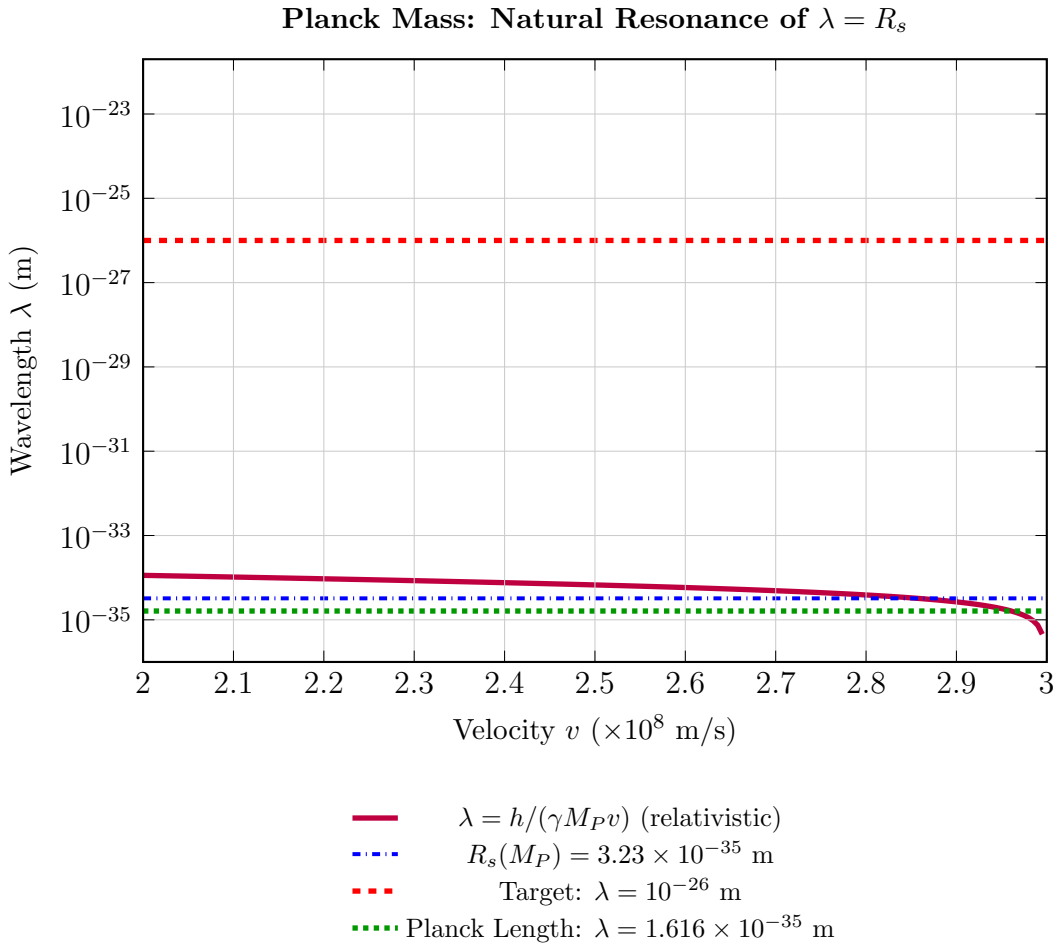


Figure 1: de Broglie wavelength of a Planck-mass particle as a function of velocity, compared to target scales.

8 Experimental Model

The experimental setup consists of a tungsten toroid with a graphene coating, tuned to achieve $\lambda \approx R_s$. The system operates in a high-vacuum, cryogenic environment to minimize interference and enhance measurement precision.

8.1 Toroid Design

- **Material**: Tungsten core, mass $M = 10\,\text{kg}$, yielding:

$$R_s = \frac{2 \cdot 6.674 \times 10^{-11} \cdot 10}{(3 \times 10^8)^2} \approx 1.48 \times 10^{-26}\,\text{m} \tag{4}$$

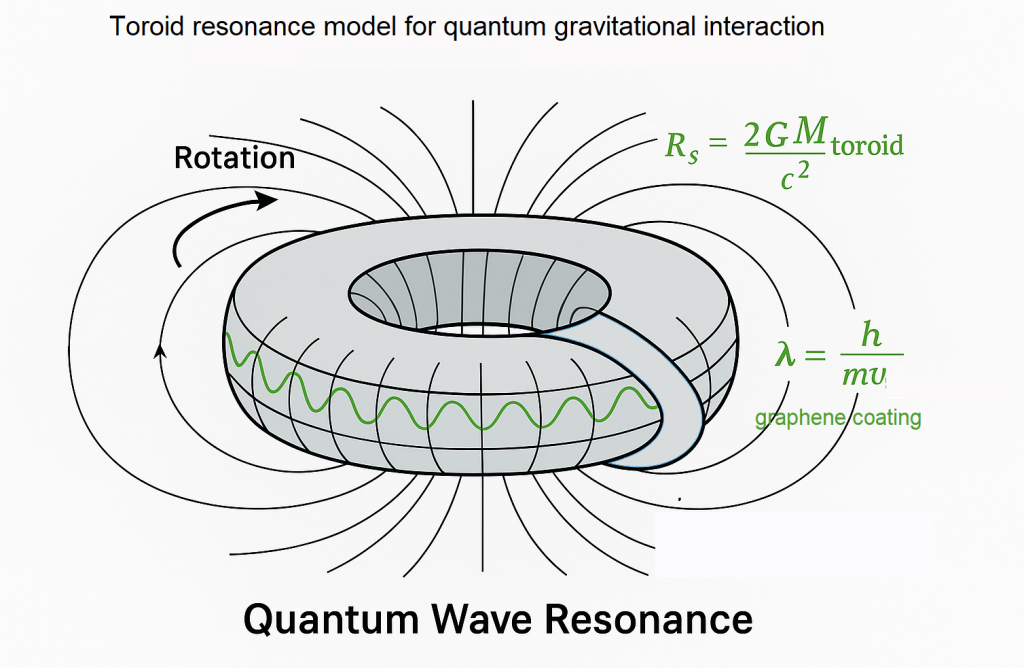


Figure 2: Toroid resonance model showing rotation, gravitational curvature, graphene coating, and the resonant condition $\lambda = \frac{h}{mv}$ aligning with $R_s = \frac{2GM}{c^2}$.

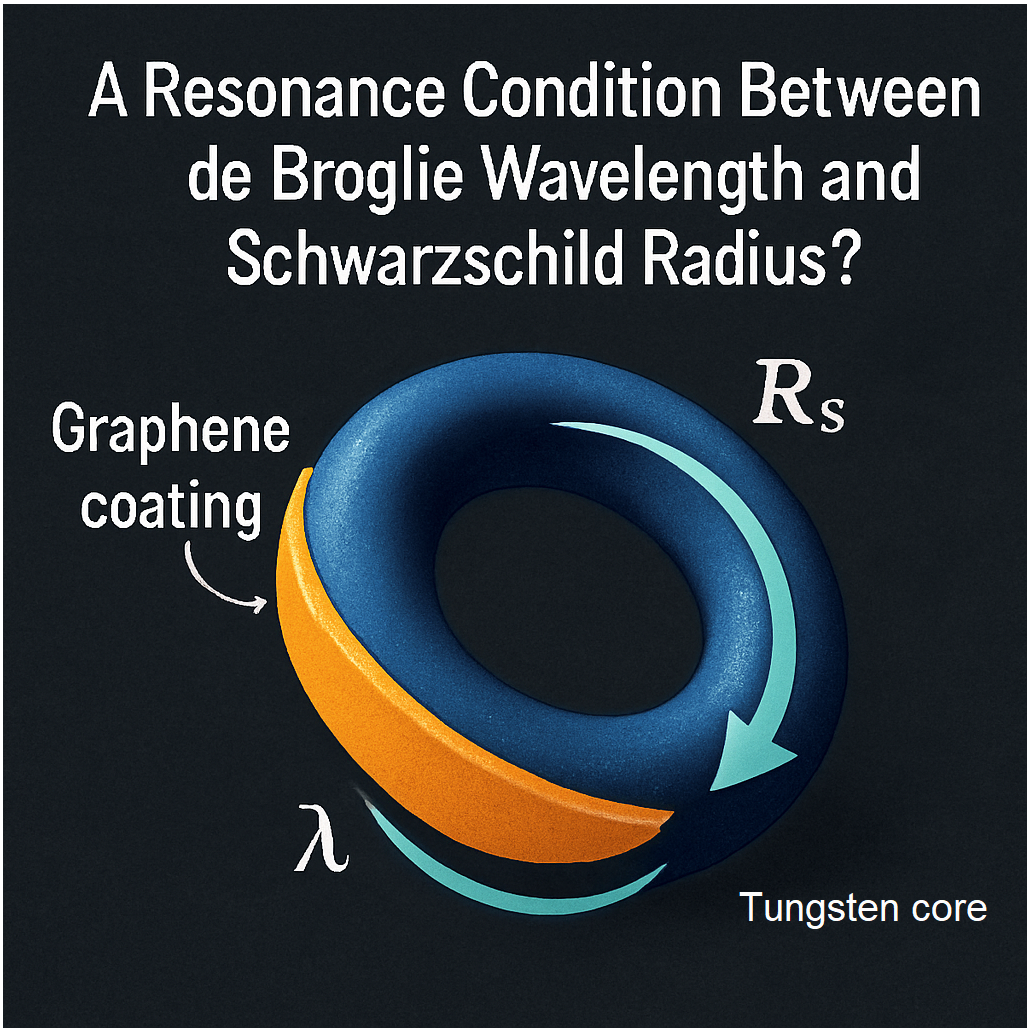


Figure 3: Toroid resonance model diagram showing tungsten core and graphene coating for the resonance $\lambda = \frac{h}{mv}$ with $R_s = \frac{2GM}{c^2}$.

- **Geometry**: Major radius $R = 0.05\,\text{m}$, tube radius $r \approx 0.023\,\text{m}$ (outer diameter $\approx$ 14.6 cm), volume $V = 2\pi^2 R r^2 \approx 520\,\text{cm}^3$ (tungsten density $19,250\,\text{kg/m}^3$). *Corrected in Version 4: the earlier tube radius of 1 cm gives only 99 cm^3, i.e. 1.9 kg.*
- **Rotation**: 500 RPM via Maxon EC-max 30 motor.

8.2 Material Selection and Justification

Tungsten was selected for the toroid core due to its exceptionally high density ($\rho = 19,250\,\text{kg/m}^3$), allowing a laboratory-sized object (e.g., 10 kg toroid) to reach a Schwarzschild radius of $R_s \approx 1.48 \times 10^{-26}$ m. This enables the curvature-generating subsystem to be both compact and massive enough to match the de Broglie wavelength of a tuned quantum carrier flow.

Tungsten also offers mechanical stability, machinability, and compatibility with cryogenic environments without requiring superconductivity. While it is not a coherent quantum medium, this is acceptable, as the gravitational curvature (R_s) and the quantum wave behavior (λ) are decoupled in this model.

For completeness, other materials and configurations were considered:

- **Superconductors**: Offer intrinsic coherence and zero resistance, ideal for generating stable quantum currents. However, they require milliKelvin cryostats and introduce magnetic shielding complications. Matching R_s would depend on the effective current mass and geometry, which are more abstract than a defined toroid.
- **Crystalline Lattices**: Provide well-defined phonon modes and vibrational coherence. But the mass contribution from individual oscillations is small, and estimating an effective λ for a bulk crystal remains ambiguous.
- **Conventional Metals**: Easily available and capable of supporting currents, but suffer from ohmic losses, lower electron mobility, and no intrinsic coherence. They do not offer substantial advantage over tungsten for R_s generation.
- **Composite or Layered Systems**: In principle, one could embed superconducting filaments inside a dense toroidal matrix, or use hybrid structures. However, these raise fabrication complexity and interpretation challenges.

Conclusion: Among these, a tungsten toroid coated with high-mobility graphene offers the clearest separation of mass and wave subsystems, allowing direct calculation of R_s from geometry and direct tuning of λ via gating and field control. This makes it an optimal configuration for initial experimental approximation of $\lambda = R_s$ resonance in a mesoscopic regime.

8.3 Graphene Subsystem

Center-of-mass convention. The de Broglie wavelength used throughout this paper for the graphene carrier subsystem is the *center-of-mass wavelength* of the collective carrier population:

$$\lambda_{\text{CM}} = \frac{h}{M_{\text{carriers}}\, v}, \tag{5}$$

where M_{carriers} is the total effective mass of all active conduction electrons in the graphene layer and v is their common drift velocity. This is the legitimate macroscopic wave-mechanical interpretation, consistent with de Broglie's original 1924 formulation, which

assigned a wavelength $\lambda = h/p$ to any moving body without requiring single-particle coherence or Bose condensation. The coherence prerequisite for observable wave behavior was elaborated later by Schrödinger, Heisenberg, and the broader quantum-mechanical program. Macroscopic phase coherence around a loop, as in Sagnac interferometry, suffices for the collective de Broglie wavelength to be physically meaningful.

- **Coating**: Single-layer graphene, surface area $\approx 453\,\mathrm{cm}^2$ (from $A = 4\pi^2 Rr$), physical mass $\approx 3.5 \times 10^{-8}\,\mathrm{kg}$.
- **Collective carrier mass**: $M_{\mathrm{carriers}} \approx 6.63 \times 10^{-14}\,\mathrm{kg}$, corresponding to $\sim 7 \times 10^{16}$ active conduction electrons counted with the free-electron mass (carrier density $\approx 1.6 \times 10^{14}\,\mathrm{cm}^{-2}$). This density is at the high end of what graphene supports; achievable with ionic-liquid gating, solid electrolyte gating, or enlarged graphene area.
- **Velocity**: Tuned to $v \approx 0.67 \times 10^6\,\mathrm{m/s}$ using a 5 GHz AC field (Keysight N5183B) and voltage source (Keithley 2635B), yielding:

$$\lambda_{\mathrm{CM}} = \frac{6.626 \times 10^{-34}}{6.63 \times 10^{-14} \cdot 0.67 \times 10^6} \approx 1.49 \times 10^{-26}\,\mathrm{m} \tag{6}$$

  matching R_s for resonance. *(Version 4: an AC field alone gives no net drift; the collective velocity must be a steady drift. See Section 8.6.)*

8.4 Measurement and Environment

- **Carrier flow**: Monitored by transport and local probes (e.g., an Omicron VT-STM). The wavelength itself, about 10^{-26} m, cannot be measured directly (see below).
- **Resonance**: Detected via current anomalies using a lock-in amplifier (Stanford Research SR860).
- **Environment**: High-vacuum (10^{-7} Torr), 4 K cryostat (Janis ST-100), mu-metal shielding.

8.5 Detection Strategy and Observable Signatures

The resonance condition $\lambda \approx R_s$ is not directly observable, as both quantities fall below current spatial resolution limits. However, indirect signatures may signal quantum–gravitational coupling, particularly through the response of the graphene subsystem and surrounding electromagnetic environment. Possible detection strategies include:

- **Conductance Shifts:** Changes in electrical conductance or carrier mobility in the graphene layer may appear under precise tuning of gate voltage and current. These could indicate altered effective mass or energy band interaction near the resonance condition.
- **Nonlinear Current Response:** The I–V characteristics may exhibit nonlinear behavior at resonance, including threshold effects or discontinuities, reminiscent of behavior in Josephson junctions or SQUIDs.
- **Frequency-locked Anomalies:** If the carrier wave is driven by a tunable AC field (e.g., 5 GHz), the system may exhibit resonance peaks or dips in the power spectral density. Lock-in amplifiers can be synchronized to track such anomalies.

- **Noise Suppression or Amplification:** Enhanced coherence at resonance may suppress thermal current noise or produce shot noise anomalies. These effects can be monitored using ultra-sensitive voltage noise spectrum analyzers.
- **Q-Factor Enhancement:** If the system behaves as a resonator, the quality factor (Q) may peak at resonance due to improved energy confinement or reduced dissipation.
- **Spin-sensitive Signatures (Optional):** Under specific configurations (e.g., spin-polarized current), one might observe tiny spin precession anomalies or altered magnetoresistance near resonance, although this requires advanced detection techniques such as NV-center magnetometry.

It is not expected that $\lambda = R_s$ matching would produce gravitational waves or singularities. Rather, the hypothesis is that resonance may subtly affect coherence, conduction, or electronic behavior—producing measurable signatures in standard laboratory apparatus. A null result remains scientifically valuable in constraining such models.

8.6 Feasibility and Main Limitation (Version 4)

Most of the equipment is standard, and the 10 kg tungsten core is practical. But one requirement was not checked in earlier versions. A collective drift of 0.67×10^6 m/s for about 7×10^{16} carriers is a current $I = Nev/(2\pi R) \approx 2.5 \times 10^4$ A around the toroid. This is far beyond what a single graphene layer can carry, and the drift velocity of graphene carriers saturates well below 10^6 m/s [12]. So the graphene design, as stated, is not feasible. It remains an illustration of the two-subsystem idea. Later papers in this series develop a superconducting ring at the Compton lock, where a small, stable supercurrent sets the condition [10, 11].

9 Toroid Resonance Analysis

The plot visualizes the collective center-of-mass de Broglie wavelength $\lambda_{\mathrm{CM}} = h/(M_{\mathrm{carriers}}\, v)$ of the graphene subsystem ($M_{\mathrm{carriers}} = 6.63 \times 10^{-14}\,\mathrm{kg}$, $\sim 7 \times 10^{16}$ electrons) against velocity v, compared to the Schwarzschild radius $R_s \approx 1.48 \times 10^{-26}\,\mathrm{m}$ of a 10 kg tungsten core and the Planck length ($\lambda = 1.616 \times 10^{-35}\,\mathrm{m}$). Resonance occurs at $\lambda_{\mathrm{CM}} \approx R_s \approx 1.5 \times 10^{-26}$ m. This is about nine orders of magnitude above the Planck length; no sub-Planck wavelength is involved (corrected in Version 4).

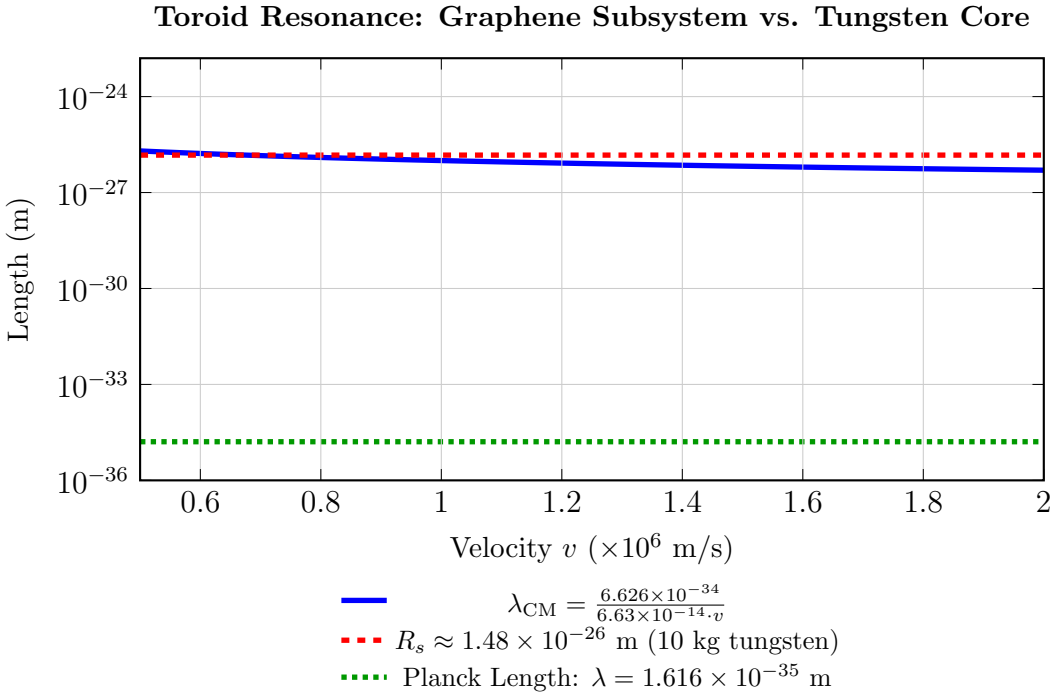


Figure 4: Graphene subsystem tuned for resonance with tungsten toroid Schwarzschild radius

10 Future Directions

The model presented here opens several paths for theoretical and experimental extension:

- Exploring other materials (e.g., topological insulators) for λ generation.
- Incorporating magnetic field interactions and spin-orbit effects in graphene.
- Designing analog systems (e.g., metamaterials) that simulate effective curvature.
- Developing indirect detection techniques for vacuum coupling or sub-wavelength resonance shifts.
- **Superconducting graphene rings (added 2026):** Subsequent work in the A-series identified intercalated and twisted-graphene superconductors (e.g., CaC_6, twisted bilayer graphene) as a tunable BEC-like platform for the broader rotational-borderline relation. In such a ring, the carriers are bound Cooper pairs rather than free electrons, and the macroscopic collective wavefunction is the single phase-coherent condensate. The lock condition (one Cooper pair per pair-Compton wavelength of the rim) can be tuned by gate voltage in twisted bilayer graphene. This represents the most engineered route currently identified for any superfluid/superconducting Compton-locked rim configuration, with a characteristic ring width of ~ 0.4 μm in CaC_6. The connection to the present $\lambda = R_s$ resonance is geometric: both rely on the same identification of the Compton wavelength with the angular-duality scale.

11 Update (2026): Matching Speeds and Further Development

This section updates the original idea with later results. It is a work in progress. Its purpose is to give the general picture and open, checkable numbers.

11.1 Any mass has a matching speed

For a single body of mass m, the de Broglie wavelength equals its own Schwarzschild radius, $\lambda = h/(m\gamma v) = R_s = 2Gm/c^2$, when

$$\gamma v = \frac{\pi M_P^2 c}{m^2}.$$

So the Planck mass is not the only case. It is the case where the match falls near the speed of light. Heavier bodies match at lower speeds. This was shown in a later paper of the series [9]. Using R_s as a length scale for masses far below stellar ones is common in the literature, from small primordial black holes [5, 6] to the mass–size chart of Carr and Rees [4].

Table U1. Each body against its own R_s.

Mass	Speed needed	R_s (m)
10 kg	4.5 nm/s	1.5×10^{-26}
1 g	0.45 m/s	1.5×10^{-30}
1 mg	450 km/s	1.5×10^{-33}
Planck mass	$0.95\,c$	3.2×10^{-35}
1 ng	$\gamma \approx 1.5 \times 10^{9}$	1.5×10^{-39}
proton	energy 5×10^{47} eV	2.5×10^{-54}

Below about half the Planck mass (about 11 μg), R_s itself is shorter than the Planck length, 1.6×10^{-35} m. The required energy also grows very fast. For a proton it is far beyond any energy reached in experiments.

11.2 A part against the whole body

In the toroid, the moving carriers, or any segment of them, are compared with R_s of the whole 10 kg body, $R_s = 1.49 \times 10^{-26}$ m. Then the match fixes the momentum of the moving part, not its mass:

$$p = \frac{h}{R_s} = 4.5 \times 10^{-8}\ \text{kg m/s}.$$

A lighter part, such as a smaller segment or a single electron, must move faster.

Table U2. A part of mass m against R_s of a 10 kg body.

Mass of the moving part (segment)	Speed needed
10 kg (whole body)	4.5 nm/s
1 g	45 μm/s
17 μg	2.6 m/s
1 μg	45 m/s
1 ng	45 km/s
1 pg	$0.15\,c$
proton or electron	energy 8×10^{19} eV

For a single proton or electron the energy is about 13 J, or 8×10^{19} eV. This is within the range of the highest-energy cosmic rays observed. It exists in nature, but not in a laboratory.

11.3 Rotation

In a rotating body the speed runs from zero on the axis to its maximum at the rim, so the matching speed can be met at some radius. A body whose density falls as $1/r^2$ from

the axis can bring every shell to the same length under one common rotation. This is developed in a separate note (Gravity Resonance, in preparation). It is work in progress, together with other directions such as the Compton lock.

11.4 Further development and status

The single-body result is developed in [9]. A different and more practical condition, a superconducting ring at the Compton lock, $IR = 2ec$, is developed in [10, 11]. This is a work in progress. The numbers above are simple and can be checked by anyone. Independent checking, criticism and collaboration are welcome. For collaboration, please contact the originator: gravity.resonance@proton.me.

Appendix A: Relativistic Consideration at Planck Mass

In the relativistic regime, the de Broglie wavelength λ can be adjusted to match the Schwarzschild radius R_s. At the Planck mass M_P the match occurs at a definite speed, derived below.

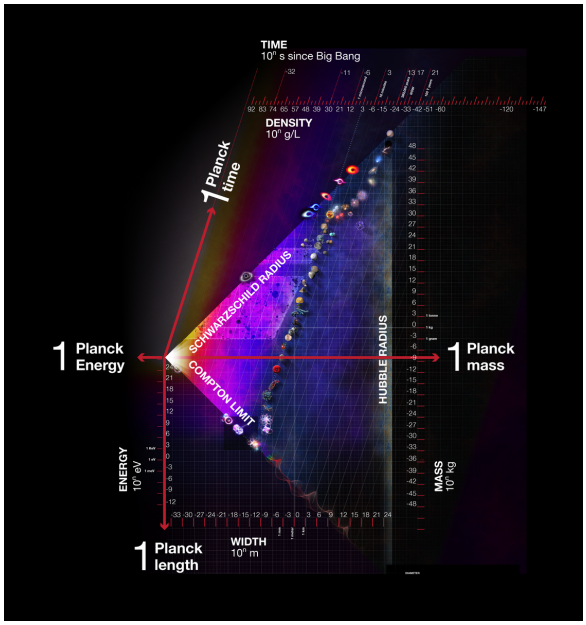


Figure 5: "Triangle of Everything" diagram showing physical systems bounded by the Compton wavelength (λ_C), Schwarzschild radius (R_s), and Planck scale, as published on Wikipedia [15].

Note: The Compton wavelength $\lambda_C = \frac{h}{mc}$ is a distinct relativistic invariant, while the de Broglie wavelength $\lambda = \frac{h}{mv}$ varies with velocity and is tunable in laboratory setups. This work focuses on the latter.

To see this explicitly, consider the relativistic momentum:

$$\lambda = \frac{h}{\gamma m v} = \frac{h\sqrt{1 - v^2/c^2}}{mv} \tag{7}$$

For the Planck mass the two lengths are equal when $\gamma v = \pi c$, that is $v \approx 0.953\, c \approx 2.86 \times 10^8$ m/s:

$$\lambda = \frac{h}{\gamma M_P v} = \frac{h}{\pi M_P c} = 2\,\ell_P = \frac{2GM_P}{c^2} = R_s \tag{8}$$

Corrected in Version 4. Earlier versions wrote $\lambda \to h/(M_P c)$ as $v \to c$; in fact $\lambda \to 0$ as $v \to c$.

This equality has been visually illustrated in the so-called *Triangle of Everything* diagram [15], where Compton wavelength λ_C, Schwarzschild radius, and Planck scale intersect. At the Planck mass the Compton wavelength, $h/(M_P c) = 2\pi\ell_P$, differs from $R_s = 2\ell_P$ by a factor π; the conceptual proximity is still important. The same crossing of the quantum and gravitational lines appears in the early mass–size chart of Carr and Rees [4].

Appendix B: Conceptual Nanoparticle Cloud Pathway

This appendix presents a speculative extension of the main toroidal model. It considers distributed ensembles of tungsten and graphene-based nanoparticles as potential configurations for exploring the de Broglie–Schwarzschild resonance condition. These ideas are retained solely as a conceptual pathway and **not** as a definitive experimental design.

Possible Configurations

- A massive tungsten core toroid with graphene-coated nanoparticles distributed on its surface.
- A heterogeneous cloud consisting of graphene-coated tungsten nanoparticles, and optionally some pure tungsten nanoparticles.
- Other combinations of materials, particle size, coating fraction, and total mass that could, in principle, tune the collective de Broglie wavelength to approach the system's Schwarzschild radius, potentially allowing λ to match even smaller R_s more easily.

Current Challenges and Limitations

- **Packing and mass effects:** The total graphene-coated surface area and conduction-electron mass complicate direct comparisons with a continuous tungsten toroid. Achieving resonance may require (i) very large device mass or volume, (ii) highly selective coherence engineering, or (iii) particle sizes outside the "nanoparticle" regime.
- **Physical interactions:** Interparticle coupling, tunneling, van der Waals aggregation, and electron–phonon decoherence introduce additional uncertainties.
- **Numerical estimates:** Previous detailed calculations for a toroidal nanoparticle cloud are now considered premature, as they do not capture the complex interactions and coherence requirements realistically.

Outlook

The nanoparticle cloud concept is presented here to illustrate a potential pathway for future research into mesoscopic systems satisfying $\lambda \approx R_s$. Future avenues may include configurations in which the total number of conduction electrons exceeds that of the original graphene sheet, or mixed arrangements of graphene-coated and pure tungsten particles.

Interested researchers are invited to refine, upgrade, and propose additional combinations in the search for resonance, exploring parameter spaces such as packing fraction, particle size, coating fraction, coherence mechanisms, and confinement strategies. Rigorous theoretical and experimental studies are needed to assess feasibility and optimize configurations.

This appendix emphasizes that the main toroidal graphene–tungsten configuration remains the primary experimentally relevant design.

Appendix C: Alternate Supercurrent-Based Toroid (Legacy Model)

A previous version of this model employed a toroid built from superconducting material (e.g., niobium or YBCO) with circulating supercurrents. This aimed to achieve $\lambda \approx R_s$ by using large toroidal mass and coherent carrier flows. However, practical limitations — including required mass, temperature control, and difficulty in tuning effective mass — led to favoring the graphene-based system. Despite its elegance, the supercurrent-based model demands either extremely low temperatures or unrealistically high mass, pushing it beyond current feasibility. Those interested in this legacy design may consult the supplementary materials.

Epilogue: The Path of Discovery

Physics, though increasingly specialized, remains a human endeavor. My contribution is that of a lifelong learner and teacher of astronomy and theoretical ideas, not of a formal research physicist. Yet the equality $\lambda = R_s$ emerged through research and curiosity—and perhaps enlightenment.

It is worth recalling that Louis de Broglie began with history before turning to physics. During World War I he worked on radio at the Eiffel Tower, far from any academic cabinet. His matter-wave proposal came in his 1924 doctoral thesis, and it *changed the course of modern physics.*

While I do not compare myself to de Broglie, I suggest that fresh ideas can—and must—emerge from outside the mainstream. This paper is invitation: for thought, collaboration, and independent experimental verification, as well as for further upgrade.

The role of tools, whether pen or algorithm, should not distract from the underlying clarity of ideas.

Should this resonance idea prove testable, it will be not because of prestige or institution, but because of clarity, persistence, and scientific openness. If it fails, let it fail honestly. If it succeeds, it may illuminate one piece of the quantum-gravity puzzle—by a simple symmetry we already hold in our textbooks.

That door remains ajar.

The shelves behind it are vast. Their books are not written in ink but in the curvature of space, the mass of electrons, and the rhythm of particles vibrating across time, in each stone and star for billions of years.

If we do not cross that threshold—out of habit, hierarchy, or hesitation—then who will?

To wait for certainty before crossing is not science—it is silence.

Let us cross. Together.

And if one day these ideas prove useful, let their light shine on everyone who dared to look—including those who built the foundations long before us, those who read the stars, those who asked strange questions, and those who, guided by intuition or faith, wrote equations on the walls of their solitude.

Let it also shine on the minds who helped refine it—both human and artificial—and the Author of the Universe whose laws, written in symmetry, still wait to be fully read.

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Acknowledgments

Earlier versions of this paper were developed with the help of ChatGPT (GPT-4, GPT-5) and Grok (Grok 3, Grok 4). Version 3 was revised with Claude Opus 4.7, and Version 4 with Claude Opus 5.5 (Anthropic). The idea and all final choices are the author's.

Citation

To cite this work, please use the following:

"Toroid Models for Quantum–Gravitational Resonance via de Broglie–Schwarzschild Symmetry", Zenodo (2025; Version 4, 2026).
DOI (all versions): 10.5281/zenodo.16371537
DOI (this version, v4): 10.5281/zenodo.23811063
Available at: https://zenodo.org/records/23811063

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